# Penguins – T-SNE Analysis

Equipo 7
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## Carga y preparación de datos

```
df <- read.csv("C:/Users/usuario/OneDrive/Desktop/Proyecto 2 Diplomado/Penguins/Club_Penguin/penguins.c
df %>% dim()
```

```
## [1] 344   8
```

```
df1 <- df[,1:8]
colSums(is.na(df1))
```

```
##           species            island    bill_length_mm     bill_depth_mm
##                 0                 0                 2                 2
## flipper_length_mm       body_mass_g               sex              year
##                 2                 2                11                 0
```

```
df1c <- df1 %>% mutate(across(where(is.numeric), ~ ifelse(is.na(.), mean(., na.rm = TRUE), .)))
colSums(is.na(df1c))
```

```
##           species            island    bill_length_mm     bill_depth_mm
##                 0                 0                 0                 0
## flipper_length_mm       body_mass_g               sex              year
##                 0                 0                11                 0
```

```
df1c <- df1c %>% mutate(
  species = as.factor(species),
  island = as.factor(island),
  sex = as.factor(sex)
)
```

# Exploración

```r
df1c %>% glimpse()
```

```
## Rows: 344
## Columns: 8
## $ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
## $ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
## $ bill_length_mm    <dbl> 39.10000, 39.50000, 40.30000, 43.92193, 36.70000, 39~
## $ bill_depth_mm     <dbl> 18.70000, 17.40000, 18.00000, 17.15117, 19.30000, 20~
## $ flipper_length_mm <dbl> 181.0000, 186.0000, 195.0000, 200.9152, 193.0000, 19~
## $ body_mass_g       <dbl> 3750.000, 3800.000, 3250.000, 4201.754, 3450.000, 36~
## $ sex               <fct> male, female, female, NA, female, male, female, male~
## $ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

```r
df1c %>% count(species)
```

```
##     species   n
## 1    Adelie 152
## 2 Chinstrap  68
## 3    Gentoo 124
```

```r
df1c %>% count(island)
```

```
##      island   n
## 1    Biscoe 168
## 2     Dream 124
## 3 Torgersen  52
```

```r
df1c %>% count(sex)
```

```
##      sex   n
## 1 female 165
## 2   male 168
## 3   <NA>  11
```

# t-SNE Básico

```r
set.seed(564)
start.time <- proc.time()
dfpeng_tsne <- df1c %>% select(where(is.numeric)) %>% scale() %>% tsne()
```

```
## sigma summary: Min. : 0.456550715184041 |1st Qu. : 0.531839696281254 |Median : 0.563246443871661 |Mea
```

```
## Epoch: Iteration #100 error is: 13.7382485560327
```

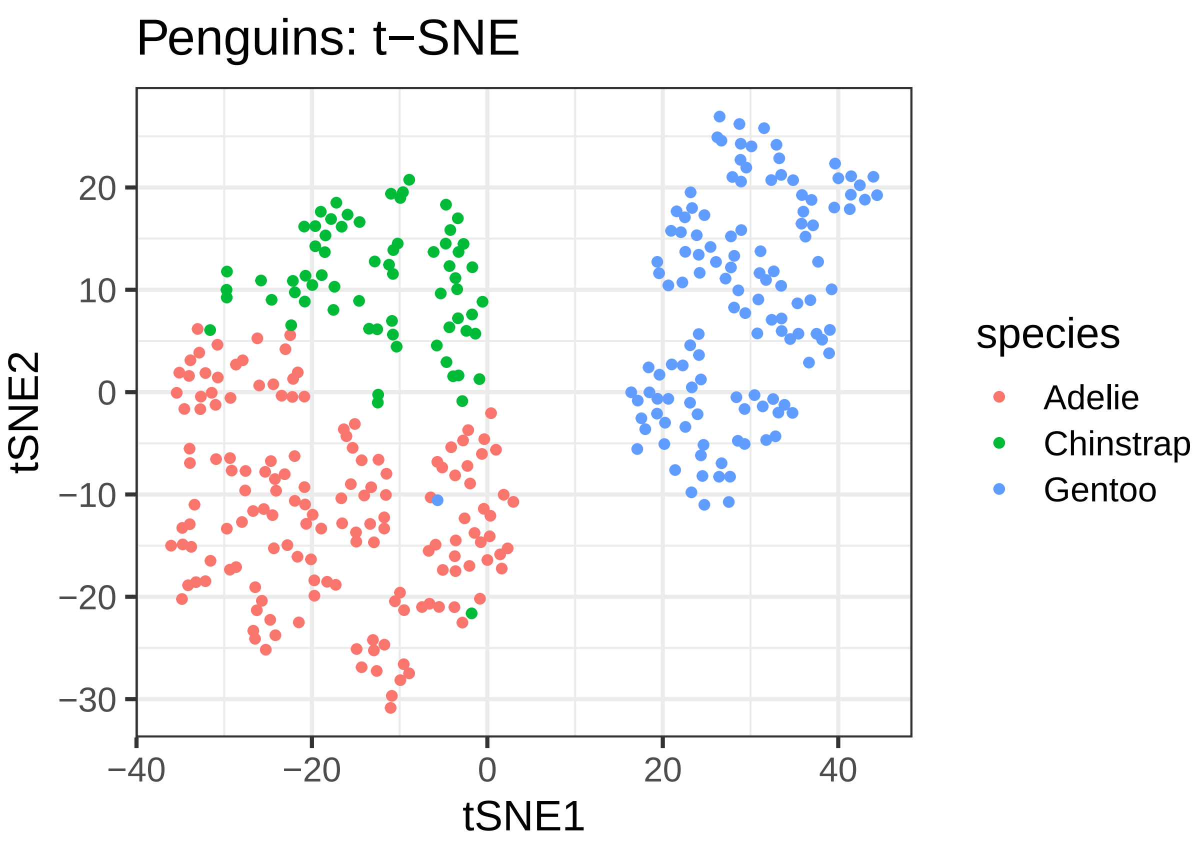
```
## Epoch: Iteration #200 error is: 0.409470427641669
```

```
## Epoch: Iteration #300 error is: 0.380563776865473

## Epoch: Iteration #400 error is: 0.375951307582076

## Epoch: Iteration #500 error is: 0.374650263808355

## Epoch: Iteration #600 error is: 0.374047206427578

## Epoch: Iteration #700 error is: 0.373702781080618

## Epoch: Iteration #800 error is: 0.373475184767524

## Epoch: Iteration #900 error is: 0.373316127889487

## Epoch: Iteration #1000 error is: 0.373200724786244
```

```
proc.time() - start.time
```

```
##    user  system elapsed
##   32.25    1.63   34.60
```

```
datos <- data.frame(tSNE1 = dfpeng_tsne[,1], tSNE2 = dfpeng_tsne[,2], species = df$species)

ggplot(datos, aes(x = tSNE1, y = tSNE2, color = species)) +
  geom_point(size = 1.1) +
  labs(title = "Penguins: t-SNE")
```

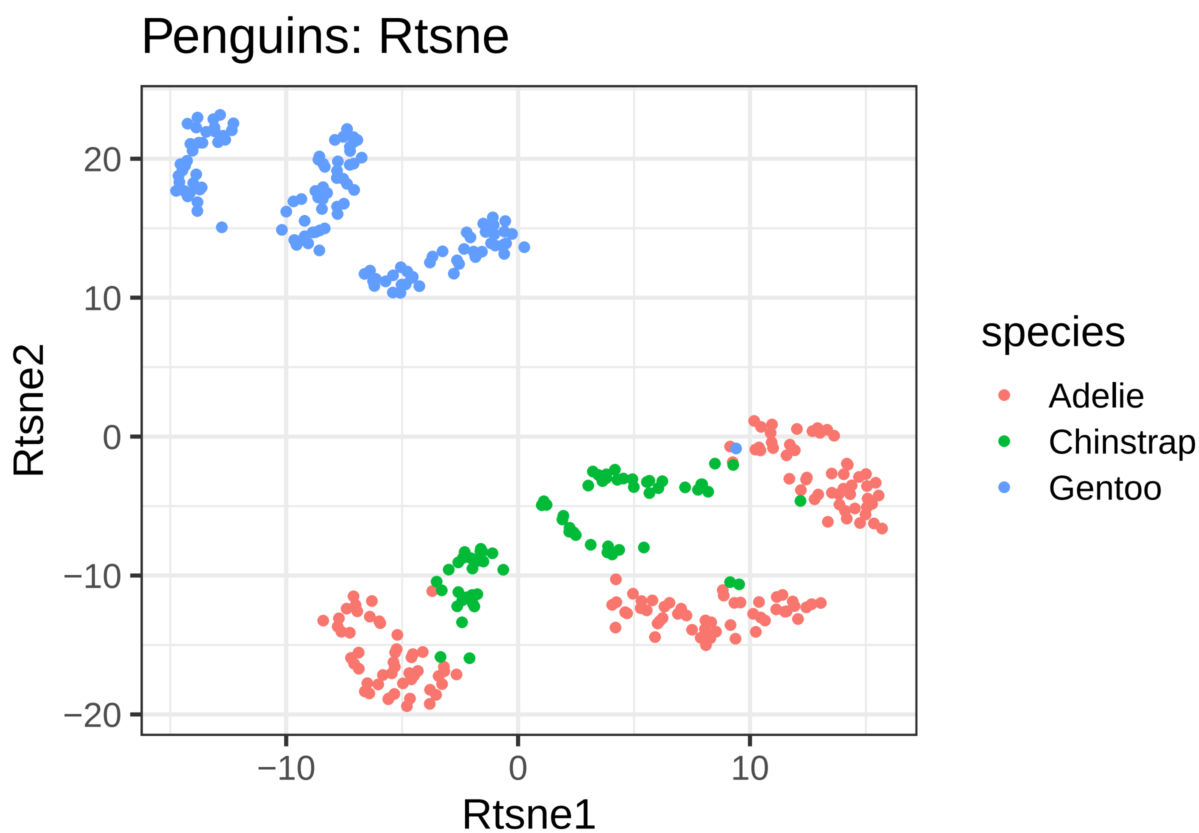


## Rtsne

```
set.seed(564)
start.time <- proc.time()
df_Rtsne <- df1c %>% select(where(is.numeric)) %>% scale() %>% Rtsne()
proc.time() - start.time
```

```
##    user  system elapsed
##    0.52    0.00    0.51
```

```
datos1 <- data.frame(Rtsne1 = df_Rtsne$Y[,1], Rtsne2 = df_Rtsne$Y[,2], species = df$species)

ggplot(datos1, aes(x = Rtsne1, y = Rtsne2, color = species)) +
  geom_point(size = 1.1) + labs(title = "Penguins: Rtsne")
```




#T-SNE 3D

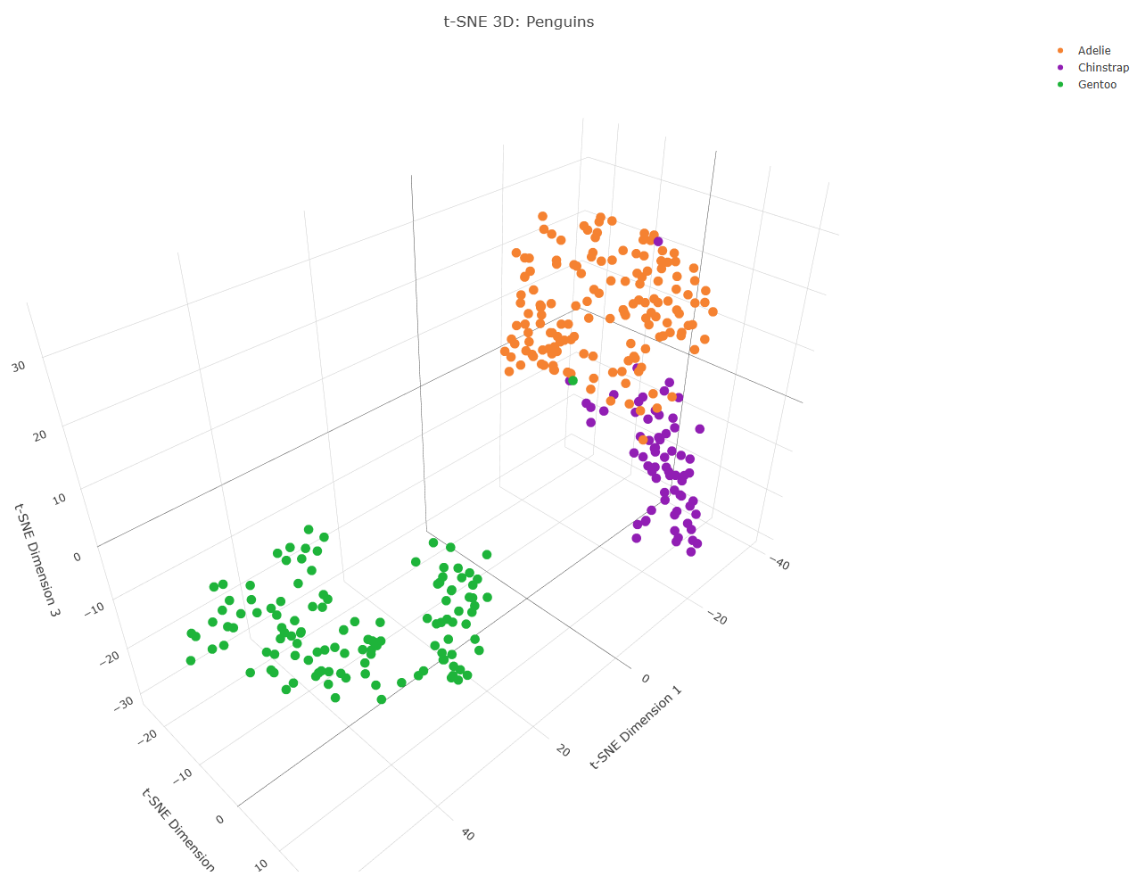
```
# ---- Run your t-SNE 3D code ----
set.seed(564)

df_tsne3 <- df1c %>%
  dplyr::select(where(is.numeric)) %>%
  scale() %>%
  tsne(k = 3)
```

```
## sigma summary: Min. : 0.456550715184041 |1st Qu. : 0.531839696281254 |Median : 0.563246443871661 |Me

## Epoch: Iteration #100 error is: 13.0001310835627

## Epoch: Iteration #200 error is: 0.330058184999922

## Epoch: Iteration #300 error is: 0.311975088827491

## Epoch: Iteration #400 error is: 0.306066488770616

## Epoch: Iteration #500 error is: 0.3036410082373

## Epoch: Iteration #600 error is: 0.302220843707391

## Epoch: Iteration #700 error is: 0.301274174746195

## Epoch: Iteration #800 error is: 0.300572030718756

## Epoch: Iteration #900 error is: 0.300041616083449

## Epoch: Iteration #1000 error is: 0.299610169422216
```

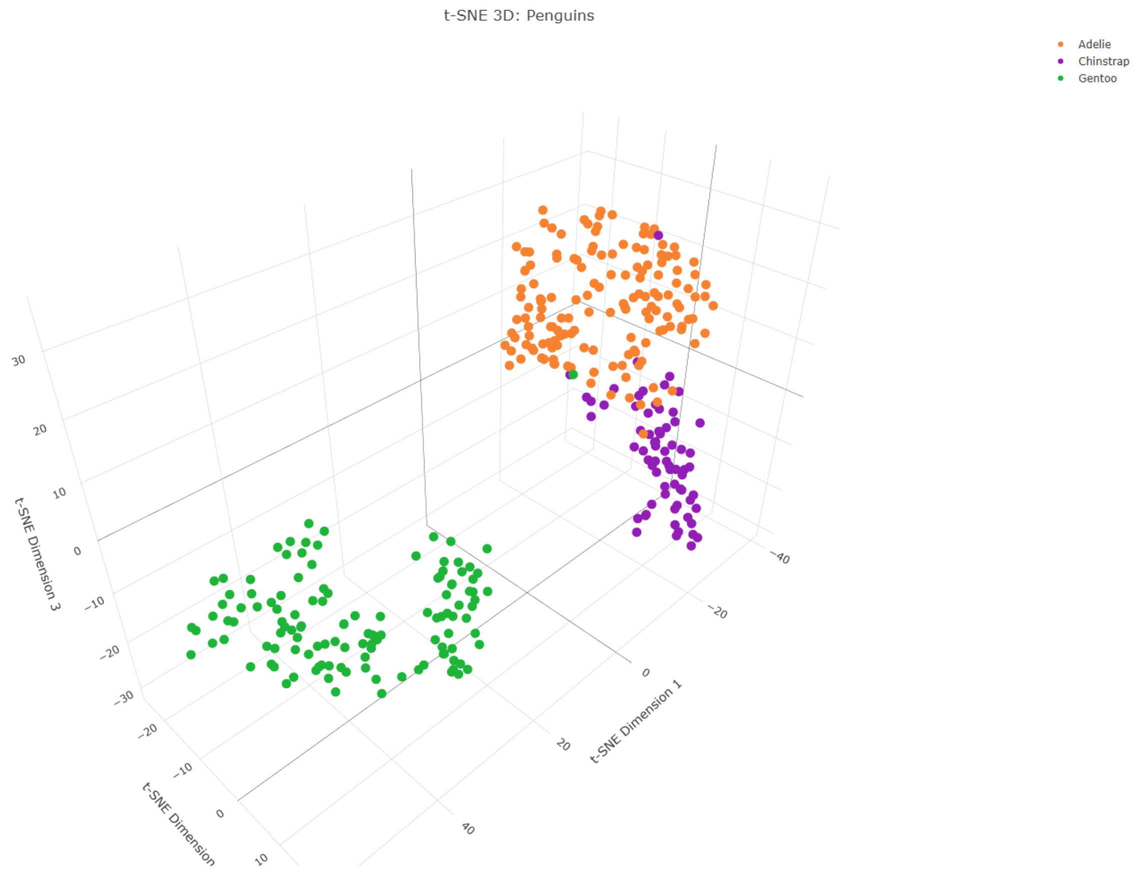
```
tsne_df3 <- data.frame(
  tSNE1 = df_tsne3[, 1],
  tSNE2 = df_tsne3[, 2],
  tSNE3 = df_tsne3[, 3],
  species = factor(df$species)
)

colors <- c("#F58231", "#911EB4", "#1eb43a")

hover_text <- paste(
  "Especie:", tsne_df3$species, "",
  "Dimension 1:", round(tsne_df3$tSNE1, 3),
  "Dimension 2:", round(tsne_df3$tSNE2, 3),
  "Dimension 3:", round(tsne_df3$tSNE3, 3)
)

# ---- Create interactive plot ----
p <- plot_ly(
  data = tsne_df3,
  x = ~tSNE1,
  y = ~tSNE2,
  z = ~tSNE3,
  type = "scatter3d",
  mode = "markers",
  marker = list(size = 6),
  text = hover_text,
  hoverinfo = "text",
  color = ~species,
  colors = colors
```

```
) %>%
  layout(
    title = "t-SNE 3D: Penguins",
    scene = list(
      xaxis = list(title = "t-SNE Dimension 1"),
      yaxis = list(title = "t-SNE Dimension 2"),
      zaxis = list(title = "t-SNE Dimension 3")
    )
  )

# ---- Save HTML version of the plot ----
htmlwidgets::saveWidget(p, "tsne3d_temp.html", selfcontained = TRUE)

# ---- Take screenshot for PDF ----
webshot2::webshot(
  "tsne3d_temp.html",
  file = "tsne3d_plot.png",
  vwidth = 1400, vheight = 1000
)
```

```
## file:///C:/Users/usuario/OneDrive/Desktop/Proyecto 2 Diplomado/Penguins/Club_Penguin/tsne3d_temp.html
```



```r
# ---- Insert screenshot into PDF ----
knitr::include_graphics("tsne3d_plot.png")
```

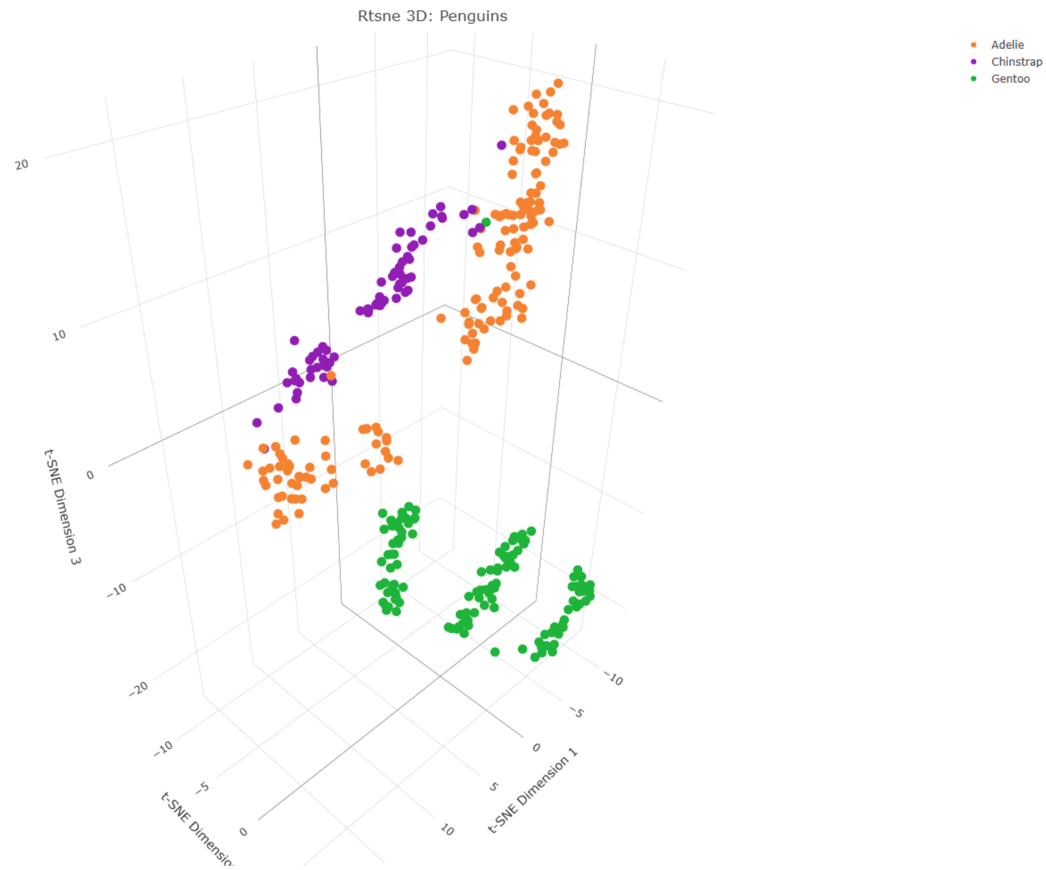



#R T-SNE 3D

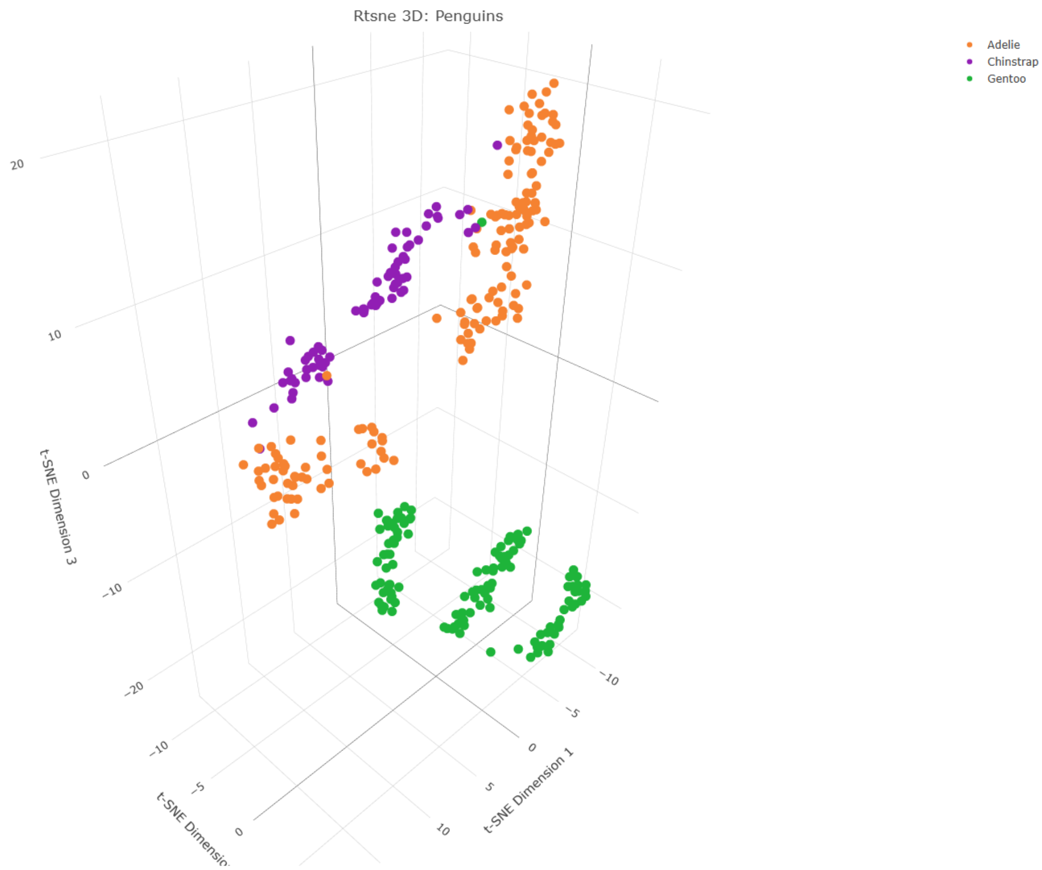
```r
library(plotly)
library(webshot2)
library(htmlwidgets)
library(knitr)
library(dplyr)

# ---- Run Rtsne 3D ----
set.seed(564)

df_Rtsne3 <- df1c %>%
  dplyr::select(where(is.numeric)) %>%
  scale() %>%
  Rtsne(dims = 3)

tsne_Rdf3 <- data.frame(
  tSNE1 = df_Rtsne3$Y[, 1],
  tSNE2 = df_Rtsne3$Y[, 2],
  tSNE3 = df_Rtsne3$Y[, 3],
```

```r
  species = factor(df$species)
)

colors <- c("#F58231", "#911EB4", "#1eb43a")

hover_text <- paste(
  "Especie:", tsne_Rdf3$species, "",
  "Dimension 1:", round(tsne_Rdf3$tSNE1, 3),
  "Dimension 2:", round(tsne_Rdf3$tSNE2, 3),
  "Dimension 3:", round(tsne_Rdf3$tSNE3, 3)
)

# ---- Create interactive 3D plot ----
p <- plot_ly(
  data = tsne_Rdf3,
  x = ~tSNE1,
  y = ~tSNE2,
  z = ~tSNE3,
  type = "scatter3d",
  mode = "markers",
  marker = list(size = 6),
  text = hover_text,
  hoverinfo = "text",
  color = ~species,
  colors = colors
) %>%
  layout(
    title = "Rtsne 3D: Penguins",
    scene = list(
      xaxis = list(title = "t-SNE Dimension 1"),
      yaxis = list(title = "t-SNE Dimension 2"),
      zaxis = list(title = "t-SNE Dimension 3")
    )
  )

# ---- Save HTML widget ----
htmlwidgets::saveWidget(p, "rtsne3d_temp.html", selfcontained = TRUE)

# ---- Screenshot for PDF ----
webshot2::webshot(
  "rtsne3d_temp.html",
  file = "rtsne3d_plot.png",
  vwidth = 1400, vheight = 1000
)
```

```
## file:///C:/Users/usuario/OneDrive/Desktop/Proyecto 2 Diplomado/Penguins/Club_Penguin/rtsne3d_temp.htm
```




```r
# ---- Insert screenshot into PDF ----
knitr::include_graphics("rtsne3d_plot.png")
```




## Exploración de parámetros t-SNE

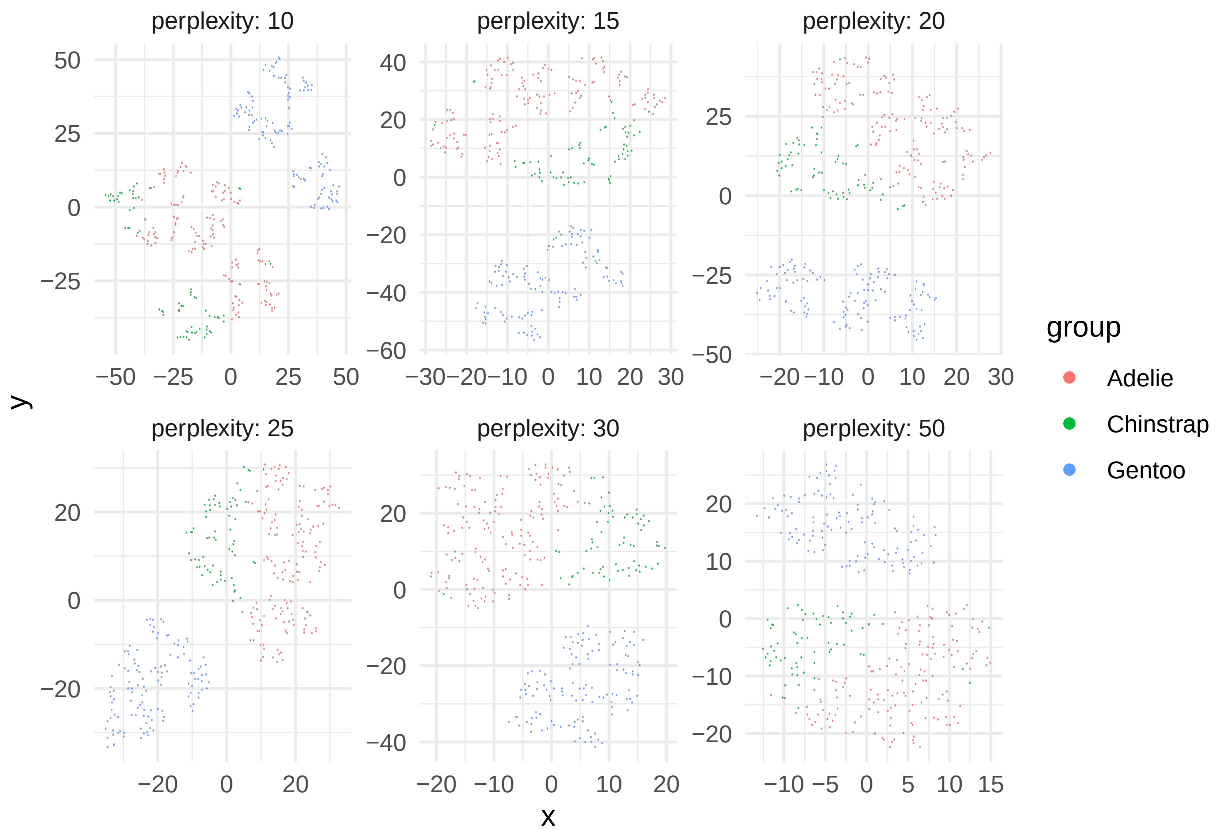
```
tsne_params <- expand.grid(perplexity = c(10,15,20,25,30,50))

numeric_df <- df1c %>% select(where(is.numeric))

set.seed(564)
tsne_res <- lapply(seq(nrow(tsne_params)), function(i) {
  tsne::tsne(X = numeric_df, max_iter = 500, perplexity = tsne_params[[1]][i])
})
```

```
## sigma summary: Min. : 0.324969406633563 |1st Qu. : 0.423069128687585 |Median : 0.45486354219507 |Mean

## Epoch: Iteration #100 error is: 14.6045496141124

## Epoch: Iteration #200 error is: 0.545540150613625

## Epoch: Iteration #300 error is: 0.460937144748885

## Epoch: Iteration #400 error is: 0.435539865397259
```

```
## Epoch: Iteration #500 error is: 0.42645893916384
## sigma summary: Min. : 0.377880463515491 |1st Qu. : 0.461865209450073 |Median : 0.493812205230027 |Mea
## Epoch: Iteration #100 error is: 14.1976294281614
## Epoch: Iteration #200 error is: 0.499081042722013
## Epoch: Iteration #300 error is: 0.437140348292237
## Epoch: Iteration #400 error is: 0.41774422601676
## Epoch: Iteration #500 error is: 0.411063278721873
## sigma summary: Min. : 0.412268794279748 |1st Qu. : 0.490690301386194 |Median : 0.523159275591973 |Mea
## Epoch: Iteration #100 error is: 14.0681644422537
## Epoch: Iteration #200 error is: 0.447155589391615
## Epoch: Iteration #300 error is: 0.40730435579213
## Epoch: Iteration #400 error is: 0.395997285029202
## Epoch: Iteration #500 error is: 0.392047624446419
## sigma summary: Min. : 0.438363876332315 |1st Qu. : 0.513432937859406 |Median : 0.544528044108326 |Mea
## Epoch: Iteration #100 error is: 13.9715443537125
## Epoch: Iteration #200 error is: 0.438374670900437
## Epoch: Iteration #300 error is: 0.403605897175491
## Epoch: Iteration #400 error is: 0.396222141141037
## Epoch: Iteration #500 error is: 0.39393131800236
## sigma summary: Min. : 0.456550715184041 |1st Qu. : 0.531839696281254 |Median : 0.563246443871661 |Mea
## Epoch: Iteration #100 error is: 13.4992831540457
## Epoch: Iteration #200 error is: 0.404599712927645
## Epoch: Iteration #300 error is: 0.383822832340434
## Epoch: Iteration #400 error is: 0.37923432367521
```

```
## Epoch: Iteration #500 error is: 0.377925574968271

## sigma summary: Min. : 0.499203681016714 |1st Qu. : 0.589775681524703 |Median : 0.620393652541271 |Mea

## Epoch: Iteration #100 error is: 13.0434806653928

## Epoch: Iteration #200 error is: 0.390919493983002

## Epoch: Iteration #300 error is: 0.386160700211491

## Epoch: Iteration #400 error is: 0.385768547651697

## Epoch: Iteration #500 error is: 0.385732096137412
```

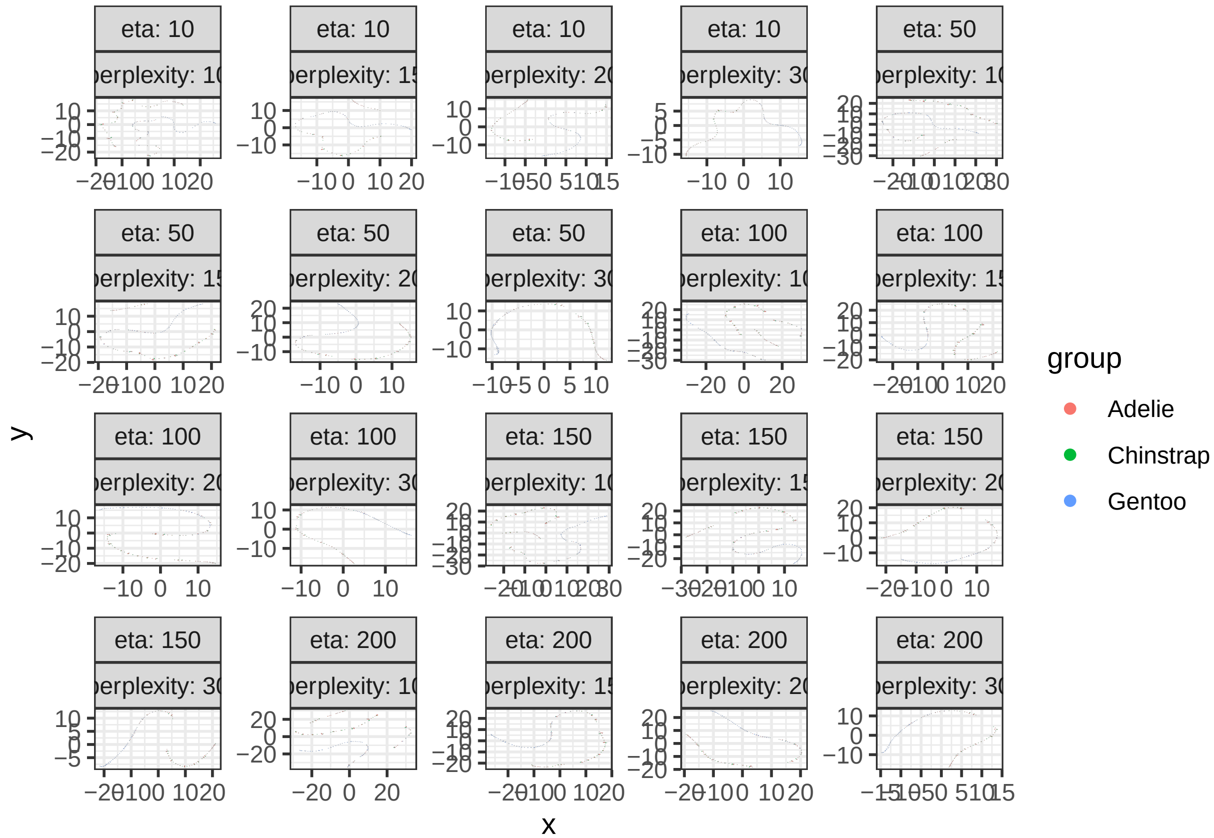
```r
d1 <- rbindlist(lapply(seq(nrow(tsne_params)), function(i) {
  data.table(
    x = tsne_res[[i]][,1],
    y = tsne_res[[i]][,2],
    perplexity = tsne_params[[1]][i],
    group = df$species
  )
}))

ggplot(d1) +
  geom_scattermore(aes(x = x, y = y, colour = group), pointsize = 2) +
  facet_wrap("perplexity", labeller = label_both, scales = "free") +
  theme_minimal()
```



## Exploración Rtsne (perplexity y eta)

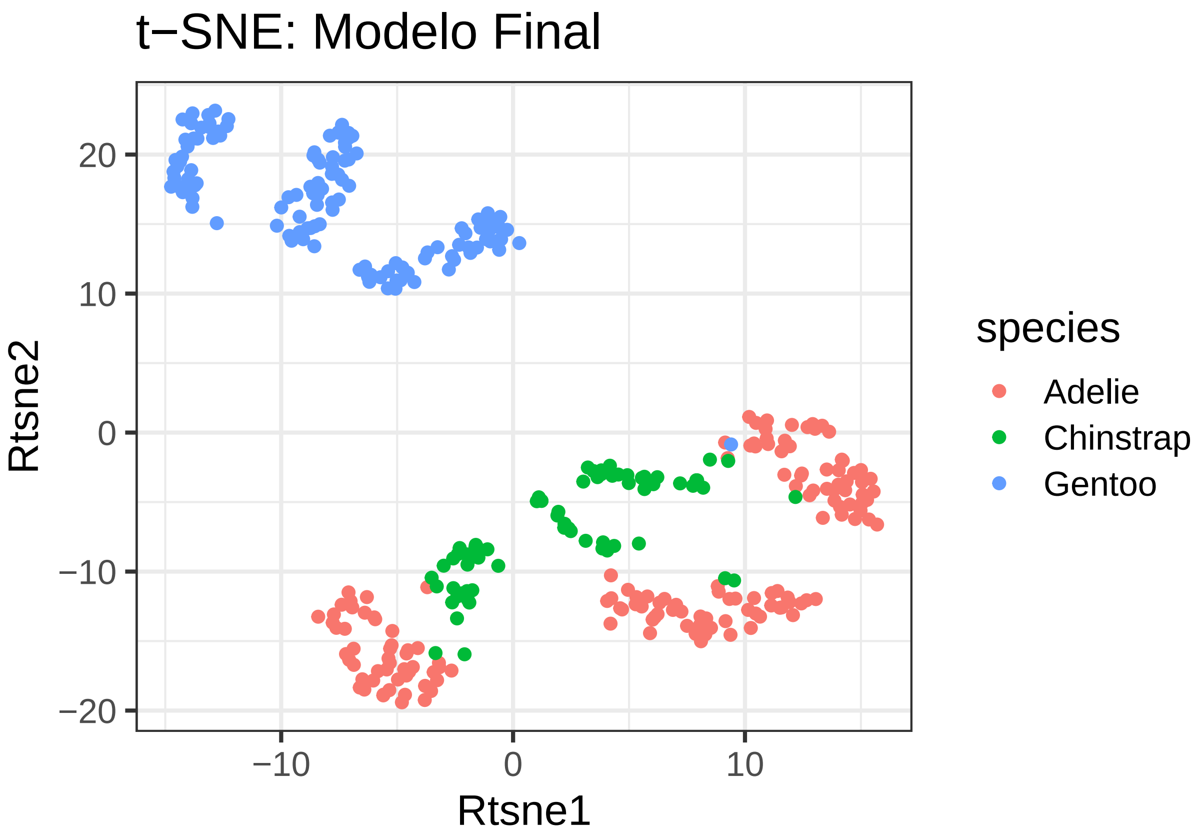
```
set.seed(564)
Rtsne_params3 <- expand.grid(perplexity=c(10,15,20,30), eta=c(10,50,100,150,200))

Rtsne_res3 <- lapply(seq(nrow(Rtsne_params3)), function(i) {
  Rtsne(
    X = numeric_df,
    max_iter = 500,
    verbose = FALSE,
    perplexity = Rtsne_params3$perplexity[i],
    eta = Rtsne_params3$eta[i],
    check_duplicates = FALSE,
    pca = FALSE
  )
})
```

```
d4 <- rbindlist(lapply(seq(nrow(Rtsne_params3)), function(i) {
  data.table(
    x = Rtsne_res3[[i]]$Y[,1],
    y = Rtsne_res3[[i]]$Y[,2],
    perplexity = Rtsne_params3[[1]][i],
    eta = Rtsne_params3[[2]][i],
    group = df$species
  )
}))

ggplot(d4) +
  geom_scattermore(aes(x = x, y = y, colour = group), pointsize = 2) +
  facet_wrap(eta~perplexity, labeller = label_both, scales = "free", ncol = 5) +
  theme_bw()
```




## Modelo Final

```r
datos2 <- data.frame(Rtsne1 = df_Rtsne$Y[,1], Rtsne2 = df_Rtsne$Y[,2], species = df$species)

ggplot(datos2, aes(x = Rtsne1, y = Rtsne2, color = species)) +
  geom_point(size = 1.5) +
  labs(title = "t-SNE: Modelo Final")
```


t−SNE: Modelo Final
Rtsne2
Rtsne1
20
10
0
−10
−20
−10
0
10
species
Adelie
Chinstrap
Gentoo